Rigorous Geometrical Derivation of the Area of Parallelogram $ABCD$

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1 Problem Statement

Let $ABCD$ be a parallelogram with $\angle BAD < 90^\circ$. A circle is tangent to sides DA, AB, and BC, intersecting the diagonal AC at points P and Q such that $AP < AQ$. Given that $AP = 3$, $PQ = 9$, and $QC = 16$, we find the exact area of the parallelogram $ABCD$.

2 Step 1: Application of Power of a Point

Let the circle be tangent to side AB at point Z and tangent to side BC at point Y. The diagonal AC acts as a secant line passing through the circle at points P and Q. We calculate the total lengths along this secant line from the vertices:

$$\begin{aligned}
&AP = 3 \\
&AQ = AP + PQ = 3 + 9 = 12 \\
&PC = PQ + QC = 9 + 16 = 25 \\
&QC = 16
\end{aligned}$$

By the Tangent-Secant theorem (Power of a Point) from vertex A:

$$AZ^2 = AP \cdot AQ = 3 \cdot 12 = 36 \implies AZ = 6 \tag{1}$$

Similarly, by the Power of a Point from vertex C:

$$CY^2 = CQ \cdot CP = 16 \cdot 25 = 400 \implies CY = 20 \tag{2}$$

3 Step 2: Projections and Parallelogram Height

Let the circle be tangent to the bottom side AD at point X. Because the circle is simultaneously tangent to the parallel lines AD and BC, the vertical distance between the points of tangency X and Y corresponds exactly to the vertical height h of the parallelogram.

Let us drop a perpendicular altitude from vertex C down to the line containing AD, meeting it at point R. The horizontal shift from the tangency point X to Y is preserved under parallel translation, which implies that the horizontal baseline distance AR is the sum of the non-overlapping tangent segments from A and C:

$$AR = AX + CY \tag{3}$$

Since tangent segments from a single point to a circle are equal, we know $AX = AZ = 6$. Therefore:

$$AR = 6 + 20 = 26 \tag{4}$$

We consider the right-angled triangle $\triangle ARC$, where the hypotenuse is the full diagonal $AC = 3 + 9 + 16 = 28$. Applying the Pythagorean theorem yields the height $h = CR$:

$$\begin{aligned} CR &= \sqrt{AC^2 - AR^2} \\ h &= \sqrt{28^2 - 26^2} = \sqrt{784 - 676} = \sqrt{108} = 6\sqrt{3} \end{aligned}$$

4 Step 3: Determining the Free Tangent Segment

Let the shared tangent lengths from vertex B to the circle be $BZ = BY = t$. The side lengths of the parallelogram are expressed as:

$$\begin{aligned} AB &= AZ + ZB = 6 + t \\ BC &= BY + YC = t + 20 \end{aligned}$$

Let us construct a right triangle under the slant side AB by dropping a vertical line from B to AD. The height of this triangle is $h = 6\sqrt{3}$. Because the horizontal baseline distance from A to the circle's center is 6, and the horizontal baseline distance from B back to the circle's center is t, the horizontal base of this slant right triangle is given by $|6 - t|$.

Applying the Pythagorean theorem to this configuration:

$$\begin{aligned} (6\sqrt{3})^2 + (6 - t)^2 &= (6 + t)^2 \\ 108 + (36 - 12t + t^2) &= 36 + 12t + t^2 \end{aligned}$$

Canceling out 36 and t^2 from both sides simplifies the equation linearly:

$$\begin{aligned} 108 - 12t &= 12t \\ 24t &= 108 \implies t = \frac{108}{24} = \frac{9}{2} = 4.5 \end{aligned}$$

5 Step 4: Final Area Calculation

With $t = 4.5$, we find the exact dimensions of the base (BC) and the altitude (h) of the parallelogram:

$$\begin{aligned} \text{Base } (BC) &= 4.5 + 20 = 24.5 = \frac{49}{2} \\ \text{Height } (h) &= 6\sqrt{3} \end{aligned}$$

The total area $\mathcal{A}$ of the parallelogram $ABCD$ is computed as:

$$\mathcal{A} = \text{Base} \times \text{Height} = \left(\frac{49}{2}\right) \times 6\sqrt{3} = 147\sqrt{3} \tag{5}$$

Thus, the area is expressed in the requested form $m\sqrt{n}$ where $m = 147$ and $n = 3$. The final computation yields:

$$m + n = 147 + 3 = 150$$